OPTIMA ACADEMY ONLINE · 2026–2027

Course Syllabus

Digital Discoveries · Grade 7 Computer Science

1 · Course Identity

Field	Detail
Course	Digital Discoveries (M/J Grade 7 Digital Discoveries, #0200022)
Grade / type	7 · Elective · Level 2
Length	Full school year (~38 instructional weeks, 4 quarters)
Delivery	Fully online and asynchronous — no live meetings
School year	2026–2027 (first day Aug 12, 2026 · last day May 28, 2027)
Facilitator	[to be completed by teacher]

Pacing note (for the registrar / authorizer)

The state lists course #0200022 with a length of “Semester.” Optima delivers it as a **full year** — a deliberate, deeper-treatment implementation with more inquiry, discussion, and a capstone portfolio. Please confirm how full-year delivery is coded on the master schedule and transcript.

2 · Course Description

Digital Discoveries introduces students to computer science as a vehicle for problem-solving, communication, and personal expression. The course focuses on the visible aspects of computing and encourages students to see where computer science exists around them and how they can engage with it as a tool for exploration and expression. In Optima’s classical tradition, we treat computing as the discipline of ordered thought.

3 · The Classical Framing

The year is structured by the **Trivium**: **Grammar** (foundations — what computing is, hardware & software, tools, honest collaboration), **Logic** (reasoning — Boolean logic, operator precedence, variables, lists, tracing, iteration), and **Rhetoric** (expression — programs, simulations, web content, a capstone portfolio). Each unit opens on a primary-source “great book” of computing, and Optima’s virtues — courage, self-governance, perseverance, honesty, courtesy, service, responsibility — are woven through as recurring reflections.

4 · What You Will Learn

- **Collaboration & communication** — working well on a team, with the right tools and roles.
- **How computers work** — the relationship between hardware and software, and researching effectively online.
- **How people & machines interact** — interfaces, sensors and devices, and adaptive technologies.
- **Health, safety & citizenship** — staying well online and treating others with courtesy.
- **Programming & problem-solving** — logic, expressions, variables, lists, tracing, loops, simulations, and building real programs.
- **Responsible, ethical technology use** — and a look at the careers this field opens up.

This course meets the full Florida state requirements for Grade 7 computer science. Teachers keep a complete record of how the course meets every standard; families may request it at any time.

5 · Scope & Sequence at a Glance

Quarter 1 — Grammar (Aug 12 – Oct 9, 2026)

Unit	Anchor text
1 · The Machine & the Human	Douglas Engelbart, “The Mother of All Demos” (1968)
2 · The Honest Collaborator	Vannevar Bush, “As We May Think” (1945)

Quarter 2 — Logic (Oct 12 – Dec 18, 2026)

Unit	Anchor text
3 · The Laws of Thought	George Boole, *An Investigation of the Laws of Thought* (1854)
4 · Order & Operation	al-Khwārizmī, *al-Jabr* (c. 820)
5 · The First Programmer	Ada Lovelace, Notes on the Analytical Engine (1843)

Quarter 3 — Logic → Rhetoric (Jan 5 – Mar 5, 2027)

Unit	Anchor text
6 · The Universal Machine	Alan Turing, “Computing Machinery and Intelligence” (1950); Euclid’s GCD
7 · Patterns & Repetition	The Fibonacci sequence

Quarter 4 — Rhetoric & Capstone (Mar 6 – May 28, 2027)

Unit	Anchor text
8 · The First Bug & the Life Cycle	Grace Hopper’s 1947 “first bug” logbook
9 · Worlds in Miniature	Modeling & simulation (PhET / NetLogo)

Unit	Anchor text
10 · Weaving the Web — Capstone Portfolio	Tim Berners-Lee, “Information Management: A Proposal” (1989)

6 · Materials & Technology

This is a fully online, asynchronous course. Students need a **Chromebook or laptop** with a modern browser, headphones, and a Canvas account. Each lesson is an interactive reading completed in the browser — no installation and no personal accounts required. Discussion happens on class **discussion boards** on each student’s own schedule. The custom interactives are hosted on GitHub Pages and store no student data.

> **Note for the Canvas administrator**
> So the interactive lessons display inside Canvas, the administrator may need to allowlist the **github.io** domain (Content Security Policy / allowed iframe sources).

7 · Grading & Assessment

Category	Weight	What it includes
Asynchronous / independent work	70%	Interactive lessons & hands-on activities, coding practice, discussion-board posts, formative checks, and unit quizzes
Projects & Capstone Portfolio	30%	Unit projects (with rubrics) and the year-end digital portfolio

Assessment types: short formative checks and hands-on activities in each lesson; end-of-unit quizzes (auto-graded) and rubric-scored projects; and a cumulative **capstone portfolio** assembled across the year. Every assessment is tied to the standards it measures.

8 · Academic & AI-Use Integrity

- Your work must be **your own**; copying another’s work is a breach of honesty.
- **Cite your sources**, including code you adapt, in the format your teacher specifies.
- **AI assistants** may be used only when a teacher permits it for a task, and you must disclose what you used and how. AI may help you understand; it may not do the thinking the assignment asks of you.
- Nonsense or “word-salad” submissions are not a sincere attempt and earn no credit.

9 · Accommodations & Accessibility

Federal and state legislation requires the provision of accommodations for students with disabilities as identified on the secondary student's Individual Educational Plan (IEP) or 504 plan or postsecondary student's accommodations' plan to meet individual needs and ensure equal access. Accommodations change the way the student is instructed. Students with disabilities may need accommodations in such areas as instructional methods and materials, assignments and assessments, time demands and schedules, learning environment, assistive technology and special communication systems. Documentation of the accommodations requested and provided should be maintained in a confidential file.

Built-in accessibility: materials follow WCAG-minded practices — logical headings, alt text, color contrast of at least 4.5:1, keyboard-operable and screen-reader-labeled interactives, and captions or transcripts for media. Adaptive technologies are themselves a subject of study in Unit 1.

10 · Expectations & the Virtues

- **Courtesy & service** — respect everyone you work with online; help peers and receive help graciously.
- **Responsibility & self-governance** — engage actively and keep a steady pace.
- **Perseverance** — a bug is an invitation, not a defeat.
- **Honesty** — submit your own sincere work; cite what you borrow; disclose any tools.
- **Courage** — ask questions, share unfinished thinking, and present your work.

For parents & guardians

Optima's model is parent-supported. You can help by reviewing your student's end-of-unit reflections, helping them keep a steady pace in this self-paced course, and contacting the facilitator with questions. Course updates are posted in Canvas; school-wide updates come through the Homeroom Canvas course.